WASIM AKHTAR

Bengaluru, India | Open to Remote | wasimakhwork@gmail.com | Portfolio | GitHub | LinkedIn

SUMMARY

Computer Science undergraduate (AI & ML) interested in systems programming, AI systems, and developer tools. Built and open-sourced developer tools including Luna, an AI terminal shell in Rust, and Compass, a platform that matches developers to GitHub issues based on their skills and experience through deterministic ranking and bounded LLM reasoning. Seeking software engineering, AI systems, or developer tools internship opportunities.

TECHNICAL SKILLS

Languages: Rust, Python, JavaScript, TypeScript.
Frameworks: React, FastAPI.
AI / ML: Large Language Models (LLMs), Reinforcement Learning (PPO), Machine Learning
Systems & Tools: SQLite, Async I/O (Tokio), REST APIs, Docker, GitHub Actions, Linux/WSL, Git/GitHub
Libraries: NumPy, Pandas, Matplotlib

PROJECTS

Luna - AI-Powered Terminal Shell | Rust, SQLite, LLM APIs | Released | GitHub

- Architected and shipped a standalone terminal shell in Rust that translates natural language into shell commands across 6 interchangeable AI providers: OpenAI, Anthropic, Gemini, Groq, Ollama, and OpenRouter.
- Designed a deterministic, rule-based risk-classification engine from CRITICAL to LOW that inspects every command before execution and blocks destructive operations by default.
- Built a SQLite-backed persistent memory layer tracking command history, directory context, and recurring error patterns across sessions to dynamically re-rank AI suggestions per user.
- Implemented an automatic error-recovery pipeline that detects failed commands, generates AI-suggested fixes, re-validates them through the safety engine, and presents them for one-click approval.
- Developed a pattern-learning module that detects repeated command sequences and converts them into reusable, replayable workflows, minimizing manual command entry.
- Shipped end-to-end across 8 build phases with a one-line install script and zero telemetry by design.

Compass - AI Contribution Matching Platform | React, TypeScript, FastAPI, Groq API | Open Source | Live | GitHub

- Built a full-stack platform that matches developers to GitHub issues based on their skills and experience through deterministic ranking and bounded LLM reasoning.
- Solved a file-path hallucination failure mode by cross-referencing LLM output against real GitHub repository structure, ensuring every suggested file reference is verifiably accurate.
- Containerized the full stack with Docker and shipped a GitHub Actions CI pipeline running automated pytest coverage on every push; deployed live via Render for the backend and Netlify for the frontend.
- Designed the system so the LLM is scoped to exactly two responsibilities: natural-language reasoning and explanation, while all ranking-critical logic remains deterministic and inspectable.

LexRAG | Python, LangGraph, Pinecone, FastAPI, | GitHub

- Built a document Q&A; API with FastAPI, LangGraph StateGraph, Pinecone, Gemini embeddings, and Groq for grounded answers with source citations.
- Implemented ingestion, vector retrieval, relevance grading with a good/bad branch.

GitHub Analyzer Web App | JavaScript, HTML/CSS, REST APIs | Completed Project | Live

- Developed a web application that fetches and visualizes GitHub user statistics, including repository data and contribution graphs, using the GitHub REST API.
- Designed and implemented a custom e-ink/newspaper-style UI focused on readability and a distinctive retro user experience.

RESEARCH EXPERIENCE

Student Researcher - Reinforcement Learning & Optimization | Presidency University | 2025 - Present

- Developed a PPO-based reinforcement learning framework for adaptive feature-weight optimization, replacing static MCDM weighting approaches used in prior work.
- Modeled landing-site suitability on a 5058 x 5058 lunar surface grid for autonomous exploration simulations.
- Achieved an 18-22% improvement in prediction stability and consistency over baseline methods during experimentation.
- Generated 60m-resolution suitability maps used as primary inputs for downstream simulation work.
- Currently co-authoring an academic paper detailing the methodology and experimental results under faculty guidance.

EDUCATION

Presidency University | B.Tech, Computer Science & Technology (AI & ML Specialization) | 2023 - 2027 (Expected)
Relevant Coursework: Machine Learning, Artificial Intelligence, Reinforcement Learning, Blockchain Technologies

LANGUAGES

English (Fluent), Hindi (Fluent)